VIDHAN CHIRAG PRAJAPATI

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EDUCATION

University of Florida, Gainesville, FL — Masters of Computer Science (4-semester program) 2026 – 2028 (Expected)
Focus: AI for Biomedical and Environmental Health

Karnavati University, Gandhinagar, Gujarat, India — Bachelor of Technology (B.Tech) in Computer Science Aug 2022 – May 2026

TECHNICAL SKILLS

Languages: Python (Advanced), C++, C, SQL, Java, JavaScript
ML & Quant Methods: Time-series modeling and anomaly detection, ensemble methods, feature engineering, cross-validation, model evaluation (accuracy, F1), deep learning (CNN, ViT, ResNet, DenseNet), medical-image analysis
Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, pandas, MONAI, OpenCV
Data & Engineering: PostgreSQL, Git/GitHub, Linux, Flask, REST APIs, React.js, Node.js, Express.js, Jupyter, Tableau
Research Interests: Biomedical AI, Medical Imaging with Deep Learning, AI for Cybersecurity

EXPERIENCE

Machine Learning Engineer Intern — Data Vidwan, Gujarat, India (Hybrid) Nov 2025 – July 2026

- Research and develop anomaly-detection methods with senior engineers for high-frequency IoT time series (sensors, smart meters, industrial equipment), separating genuine irregularities from noise in continuously arriving, non-stationary data.
- Own the path from raw signal to model output: cleaning streaming data, engineering features, and tuning detectors on false-alarm versus missed-detection trade-offs.
- Mentor 15+ B.Tech/B.Sc students (Aspire Disruptive Skill Foundation, nonprofit NGO) through the full ML workflow in Python: data preparation, feature engineering, model training, validation, and error analysis.

Data Scientist Intern — Uniconverge Technologies Pvt. Ltd., Gujarat, India (Remote) May 2024 – Jun 2024

- Engineered a reproducible preprocessing pipeline for high-dimensional 3D brain MRI (.nii/NIfTI), decomposing volumes into axial, coronal, and sagittal slice inputs to produce consistent, model-ready feature sets.
- Benchmarked convolutional (VGG, ResNet, DenseNet) and attention-based (ViT) architectures on BRATS and TCIA, comparing accuracy and F1-score to select the strongest model on measured performance.

RESEARCH & INNOVATION

Secure Framework for Patient Safety in IoMT Using Machine Learning and ECC-AES Cryptography - IEEE LINK Published

- Karnavati University. Secured Internet-of-Medical-Things patient data by combining ML-based detection with ECC–AES encryption. Co-authors: Jay Bhuptani, Varun Jethani; Mentor: Prof. Sanjay Ajani.

Homomorphically Encrypted Graph Attention Networks for Privacy-Preserving IoMT Intrusion Detection- IEEE LINK Published

• Karnavati University. Explored privacy-preserving IoMT intrusion detection by combining Graph Attention Networks (GATs) with homomorphic encryption, enabling security analysis of sensitive medical data while preserving patient privacy. Co-authors: Varun Jethani, Jay Bhuptani, Puneet Sharma, Shailendra Narayan Singh; Mentor: Prof. Hiren Kumar Thakkar.

BIRAC E-YUVA Proposal – EarlyAid: Health Monitoring System (IoT, Fullstack Web, AI)

- Biotechnology Industry Research Assistance Council. Proposal selected for a Rs 50,000 grant. Team: Jay Bhuptani, Varun Jethani; Advisor: Dr. Puneet Sharma (Associate Dean).

PROJECTS

Elderly Health Monitoring System *(MERN, IoT, TensorFlow/PyTorch, Flask)* 2025

- Supervisor: Dr. Puneet Sharma (Associate Dean). Integrated wearable IoT sensors with a real-time MERN-based web platform.
- Implemented ensemble ML models for health-risk prediction and sleep-quality analysis, plus deep learning for skin disease and surface-level threat detection, with intuitive dashboards showing real-time alerts and predictive risk insights.

Protein Function Prediction - CAFA 6 (Kaggle) | PyTorch, ESM-2, BioPython, Gene Ontology

- Developed an ensemble of nine PyTorch multilabel classifiers using pretrained 650M-parameter ESM-2 protein embeddings, engineered amino-acid features, and taxonomy to predict protein function across three Gene Ontology categories.
- Implemented Gene Ontology hierarchy propagation and UniProt-GOA annotation integration; achieved peak individual-model validation micro-F1 scores of 0.437 for Molecular Function, 0.397 for Cellular Component, and 0.192 for Biological Process.

3D Brain MRI Glioma Classification | PyTorch, MONAI, 3D DenseNet121, NumPy

- Developed a volumetric deep learning pipeline using MONAI's 3D DenseNet121 architecture to classify high-grade versus low-grade gliomas from four-channel, 128 × 128 × 128 brain MRI volumes.
- Implemented MRI normalization, class-balanced sampling, and GPU-based model training; achieved 77.8% validation accuracy and 0.816 F1-score at the best recorded epoch across 126 validation samples.

ACHIEVEMENTS & ACTIVITIES

- **Winner, Code Unnati Innovation Marathon 2024–25** (Edunet Foundation, a CSR initiative of SAP): won a national-level marathon of 450+ student teams across four states with a scalable healthcare solution integrating IoMT, web development, and ML.
- Hackathons: Gator Quant Hacks, SwampHacks 12 , Hack This Fall, Parul University. Volunteer: Cloud Community Day 2024; WOW Ahmedabad (GDSC).

CERTIFICATIONS

- **IIT Bombay Spoken Tutorial:** Python 3.4.3 (75%); RDBMS PostgreSQL (82.5%). **Oracle Academy:** Java Fundamentals (2024).
- **EXPLIoT Academy:** Introduction to IoT Security Assessment; Medical Device Security; IoT Hardware Hacking (UART to Shell).